

Timing the steep edge: recovering picosecond time resolution from a saturated sampling calorimeter with a high-gain/low-gain ratio method

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Abstract

Future high-luminosity collider experiments require electromagnetic calorimeters that measure not only the energy but also the arrival time of showers to a few tens of picoseconds. We report time-resolution measurements made with a tungsten/LYSO:Ce sampling “shashlik” calorimeter module (RADiCAL) exposed to 25–150 GeV electrons at the CERN SPS in May 2023. The high-gain channels used for timing saturate, clipping at 820 mV and destroying the pulse peak. We recover the unclipped rising edge using a second, low-gain copy of each pulse: a constant-fraction discriminator is placed at a fixed fraction of the low-gain-predicted true peak (`hg_lgcf`), which times an edge a factor ~ 3.7 steeper than the clipped-peak foot. Using a microchannel-plate (MCP)-free differential estimator, $(DW - UP)/2$, formed from the four upstream and four downstream silicon-photomultiplier readouts, we measure the time resolution and its dependence on shower brightness. The resolution improves as the inverse of the collected light, the signature of slew-rate-limited timing, reaching 25.4 ps for the brightest, best-contained 150 GeV showers and extrapolating to a floor of 24.7 ± 1.1 ps consistent with an irreducible shower-depth fluctuation of $\sim 1 X_0$. Expressed as a reproducible amplitude threshold rather than a fixed event count, the resolution falls continuously from ~ 50 ps for all contained showers toward this floor. Across four detector builds the resolution tracks the scintillation light yield, confirming that light, not beam energy, sets the clock. We discuss the implications for detector design and a depth-corrected estimator as the route below the present floor.

Keywords: Fast timing, Sampling calorimeter, LYSO, Silicon photomultiplier, Constant-fraction discrimination, Signal saturation, 4D calorimetry

1. Introduction

At the High-Luminosity LHC and proposed future colliders, of order 100–200 simultaneous interactions per bunch crossing overlap in space. Associating energy deposits with the correct primary vertex, and rejecting out-of-time pile-up, requires that calorimeters resolve the time of a shower to a few tens of picoseconds in addition to its energy — so-called 4D, or fast-timing, calorimetry [1]. Achieving this in a radiation-hard, finely-segmented electromagnetic calorimeter is the goal of the RADiCAL R&D programme [2].

A first RADiCAL test-beam measurement reported a time resolution of ≈ 27 ps at 150 GeV using a constant-fraction discriminator at 5% of the measured (clipped) pulse height, with the resolution parametrised as $\sigma_t = a/\sqrt{E} \oplus b$ and $b = 17.5$ ps [2]. That measurement is limited by a feature of the front end: the high-gain (HG) channels optimised for timing saturate the digitiser well before the shower maximum, so the pulse peak — and with it the steepest, most informative part of the rising edge — is never recorded; nor was the saturated peak recovered.

In this paper we show that the unclipped edge can be reconstructed event by event from a simultaneously-recorded low-gain (LG) copy of the same pulse, and that timing this recovered edge improves the resolution and exposes the underlying physics that sets it. We further show that the resolution is governed by the collected scintillation light through a slew-rate law, extrapolate the irreducible floor, and identify its origin as shower-depth fluctuation.

2. Experimental setup

2.1. Detector

The module under test is a tungsten/LYSO:Ce sampling calorimeter of the “shashlik” type: alternating plates of tungsten absorber and cerium-doped lutetium–yttrium oxyorthosilicate (LYSO:Ce) scintillator, read out longitudinally by wavelength-shifting capillaries. Four capillaries traverse the stack and are each read out at both ends, giving $4 \times 2 = 8$ silicon-photomultiplier (SiPM) channels. We label the four downstream ends collectively DW and the four upstream ends UP. The transverse position of each incident electron is measured by a wire chamber and used to define a fiducial region centred on the beam spot.

Four detector builds were exposed, differing in their scintillator/absorber composition and hence in light yield:

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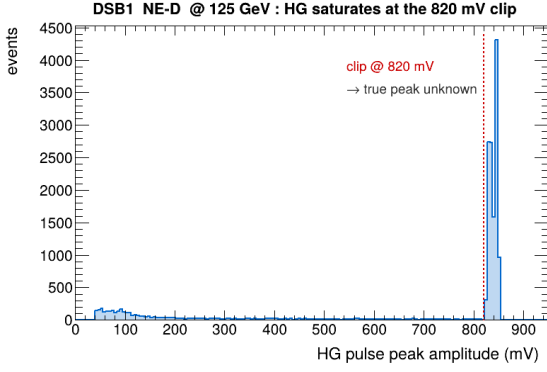


Figure 1: Distribution of recorded high-gain pulse-peak amplitudes (one channel, 125 GeV). The pile-up at ≈ 820 mV is the digitiser clip: most showers saturate, so the true peak — and the steep edge that leads to it — is not directly recorded.

a high-light LYSO build (“DSB1”), a mixed LYSO/LuAG build (“MIXED”), a low-light LuAG:Ce build (“LUAG”), and a three-segment LYSO build with an energy section (“TENENERGY”). DSB1, the brightest, is used for the headline measurements; all four are compared in Section 5.5.

2.2. Readout and digitisation

Each capillary end is read by a SiPM whose signal is split into a high-gain (HG) copy, optimised for timing, and a low-gain (LG) copy, optimised for energy linearity. All channels are digitised by DRS4 switched-capacitor waveform digitisers (CAEN DT5742) [3] sampling at 5 GS/s over a 1024-cell window, with per-cell timing and amplitude calibration applied offline. The HG channels clip at ≈ 820 mV; the LG channels remain linear across the full energy range.

2.3. Beam and timing reference

The measurements were performed in a 25–150 GeV electron beam at the CERN SPS in May 2023. A microchannel-plate photomultiplier (MCP-PMT) upstream of the module provides a common start signal; all per-channel crossing times in the reduced data are expressed relative to this MCP time. As shown below, our differential estimator cancels this common reference, so the MCP jitter does not enter the quoted resolutions.

3. Saturation and the HG/LG-ratio timing method

The difficulty is illustrated in Fig. 1: for all but the smallest showers the HG pulse rises into the 820 mV clip and piles up there, so the recorded waveform is flat across its maximum. A constant-fraction discriminator (CFD) referenced to the measured peak is then forced onto the shallow “foot” of the pulse, where the slope — and therefore the timing information — is poor.

The LG copy, although too small for precise timing on its own, remains linear and therefore measures the true

pulse amplitude that the HG channel would have reached. We exploit this with a per-end calibration

$$\text{HG}_{\text{true}} = a + b \text{LG}_{\text{peak}}, \quad (1)$$

determined from the unsaturated population (Fig. 2), which predicts the unclipped HG peak event by event. A CFD threshold is then placed at a fixed fraction (15%) of this predicted true peak and applied to the recorded HG waveform. Because the predicted peak lies far above the clip, this threshold sits on the steep, unsaturated portion of the rising edge rather than on the foot. We denote the resulting MCP-referenced crossing time hg_lgcfd .

The benefit is direct (Fig. 3): the recovered edge is timed where the local slope is 381 mV/ns, against 102 mV/ns at the clipped-peak foot — a factor ~ 3.7 . Since the slow contribution to the time resolution scales as $\sigma_t \approx \sigma_V / (dV/dt)$ with σ_V the voltage noise, this steeper edge translates directly into better timing.

4. Timing estimator and event selection

For each event we form the per-end mean crossing time from the channels with a valid pulse, and combine the two ends into the differential estimator

$$t_{\text{DW-UP}} \equiv \frac{1}{2}(\bar{t}_{\text{DW}} - \bar{t}_{\text{UP}}), \quad (2)$$

where $\bar{t}_{\text{DW}}$ and $\bar{t}_{\text{UP}}$ average up to four downstream and four upstream SiPM times respectively. The difference cancels any time common to both ends — in particular the MCP start — so $t_{\text{DW-UP}}$ is free of the reference jitter; averaging four channels per end suppresses uncorrelated single-channel noise by up to a factor of two. The complementary sum $(\text{DW} + \text{UP})/2$, which retains the common reference, is used only as a cross-check. The single-event time resolution quoted throughout is the standard deviation of a Gaussian fit to the central peak of the $t_{\text{DW-UP}}$ distribution; the small non-Gaussian shoulder from mis-measured low-amplitude channels is excluded from the fit, as in Ref. [2].

Events are required to fall within the wire-chamber fiducial region and to have a valid MCP signal. To suppress poorly-contained showers, whose low light makes them non-representative outliers, we require at least two of the four ends on each side to have a good pulse and reject events whose total low-gain signal ΣLG (our amplitude/brightness estimator) falls below 40% of the per-energy median.

A central theme of this paper is the dependence of σ_t on ΣLG . Because shower fluctuations spread the collected light at fixed beam energy, ΣLG and beam energy are distinct: the brightest events at one energy overlap the typical events at the next. We therefore analyse the resolution as a function of brightness directly.

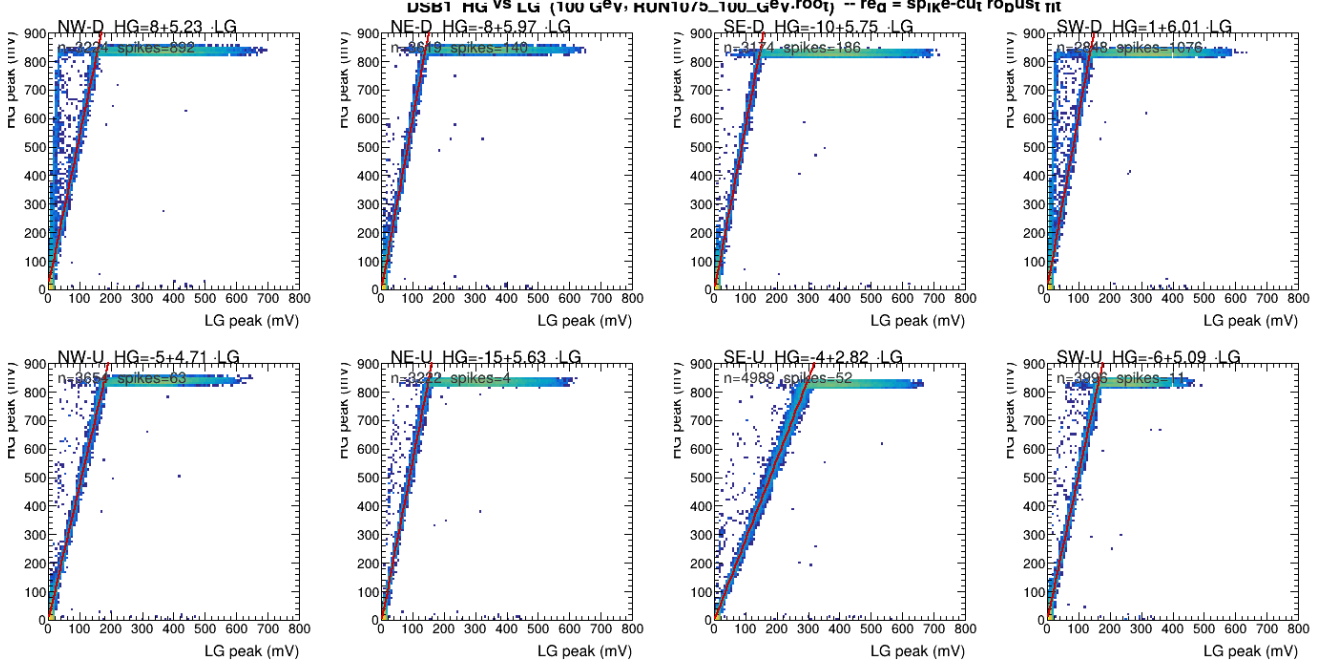


Figure 2: High-gain versus low-gain pulse-peak amplitude for the eight readout channels (100 GeV). The high-gain response saturates at 820 mV (horizontal band) while the low-gain copy remains linear; the per-channel linear fit (red line), Eq. (1), predicts the unclipped true peak event by event.

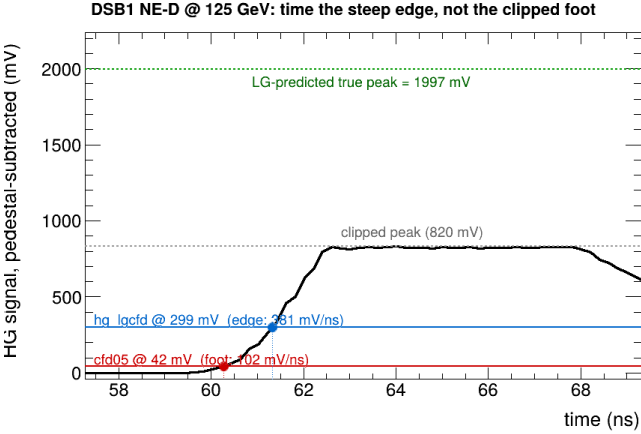


Figure 3: A single saturated high-gain waveform (one channel, 125 GeV). The recorded pulse clips at 820 mV (grey); the low-gain copy predicts a true peak of 1997 mV (green). Placing the CFD at 15% of the predicted peak (hg_lgcf, blue) times an edge of 381 mV/ns, against only 102 mV/ns at the clipped-peak foot where the 5% CFD (cfd05, red) is forced to sit — a factor 3.7 steeper.

5. Results

5.1. Timing distributions

Figure 4 shows, for DSB1 with all six energies pooled and binned by ΣLG , the t_{DW-UP} (top) and $(DW + UP)/2$ (middle) distributions together with their central-peak Gaussian fits, and the fitted σ_t versus amplitude (bottom). The distributions are clean, single-peaked and well described by the fits at every amplitude; the differential estimator is consistently narrower than the absolute one,

as expected once the common-mode reference jitter is removed. The fitted width falls monotonically with brightness.

5.2. Resolution versus brightness: the slew law

Slicing each beam energy into consecutive groups of 1000 events ordered by brightness yields the per-energy resolution curves of Fig. 5. Two features stand out. First, where the energies overlap in ΣLG they fall on a single curve: the time resolution is set by the collected light, not by the beam energy. Second, σ_t falls faster than the $1/\sqrt{\Sigma LG}$ expected from photostatistics; it scales as $1/\Sigma LG$, the signature of slew-rate-limited timing, since the edge slope is proportional to amplitude. We therefore fit

$$\sigma_t(\Sigma LG) = \sqrt{\left(\frac{a}{\Sigma LG}\right)^2 + b^2}, \quad (3)$$

to the brightest (best-measured) slice of each energy, where b is the irreducible floor as $\Sigma LG \rightarrow \infty$.

5.3. Resolution versus amplitude threshold

A fixed “brightest- N ” selection is convenient but depends on the size of the data set. The reproducible, experiment-independent quantity is the resolution of all events above an amplitude threshold. Selecting the brightest 1000 events is identical to one such cut (here $\Sigma LG > 5135$). Scanning the threshold gives the cumulative curve of Fig. 6: as the cut is raised, only brighter, better-timed showers survive and σ_t slides from ~ 50 ps for all contained 150 GeV showers toward the floor. The amplitude

DSB1 (lgcfd): per-amplitude-bin (DW-UP)/2 [top] and (DW+UP)/2 [middle] distributions, with their fitted σ_t [bottom]

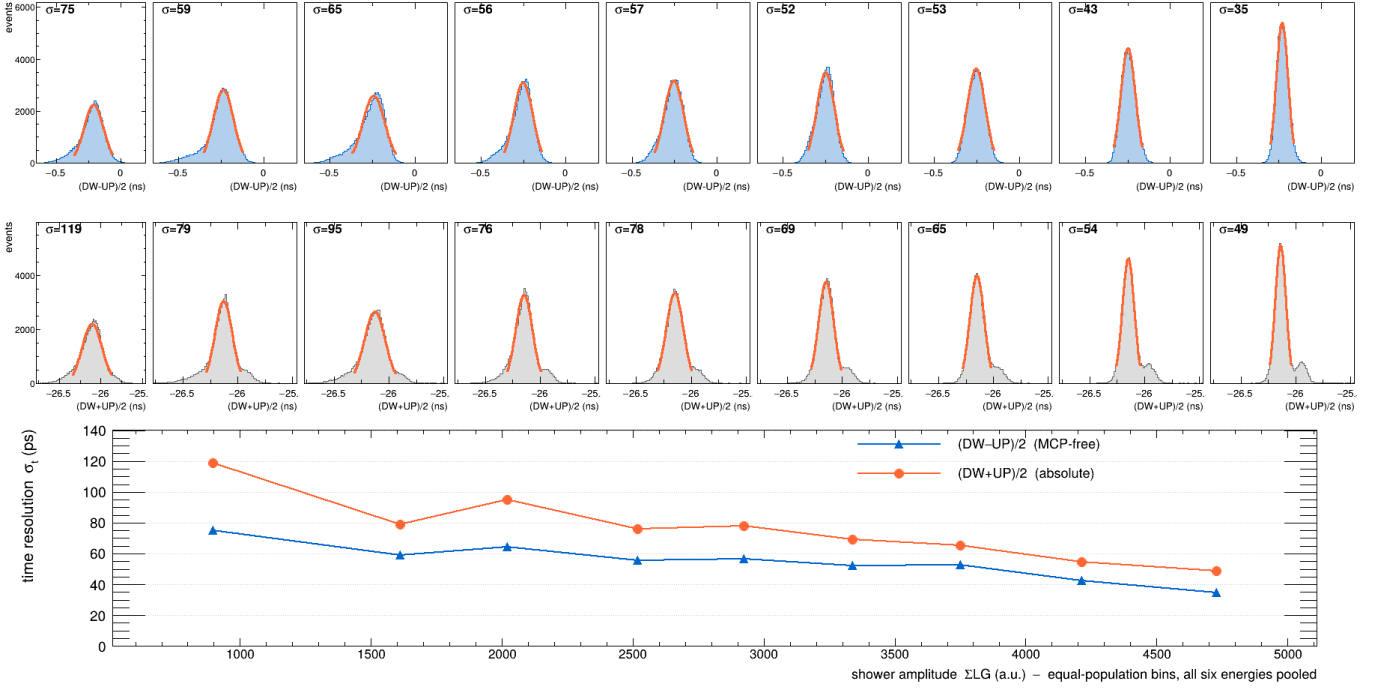


Figure 4: DSB1 timing distributions, all energies pooled and binned by amplitude ΣLG . Top: $(DW - UP)/2$ (MCP-free). Middle: $(DW + UP)/2$ (absolute). Each σ (ps) is the central-peak Gaussian fit. Bottom: the fitted σ_t versus amplitude for both estimators.

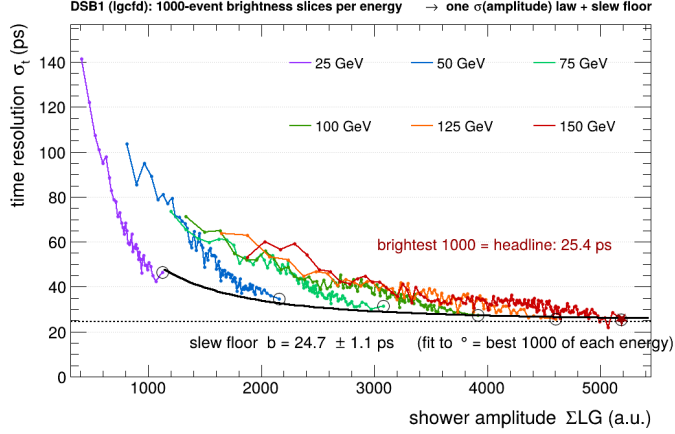


Figure 5: DSB1 resolution in consecutive 1000-event brightness slices, per beam energy. The curves collapse onto one $\sigma_t(\Sigma LG)$ law. The slew fit of Eq. (3) to the brightest slice of each energy (open circles) gives a floor $b = 24.7 \pm 1.1$ ps; the brightest 1000 of 150 GeV ($\star$) is the 25.4 ps headline.

cut alone improves the resolution by roughly a factor of two. The headline number is thus one point on a continuous curve rather than a special selection.

5.4. The floor

The slew fit, Eq. (3), yields $b = 24.7 \pm 1.1$ ps for DSB1. An independent extrapolation of σ_t versus beam energy

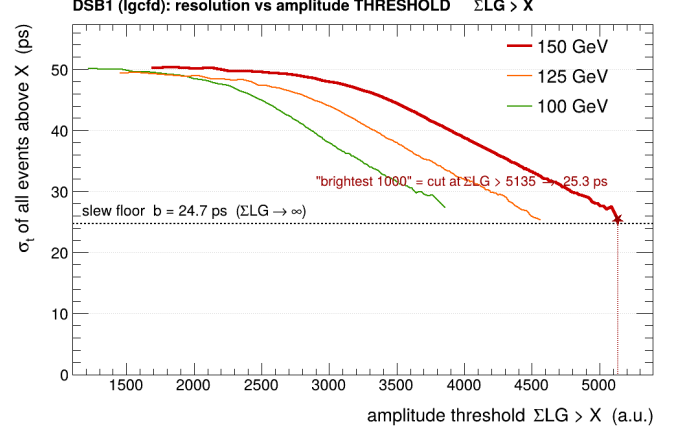


Figure 6: DSB1 resolution versus amplitude threshold: σ_t of all events with $\Sigma LG > X$, per energy. Raising the cut keeps brighter showers and the resolution approaches the floor. The “brightest 1000” headline is the single point at $\Sigma LG > 5135$ ($\star$).

gives a consistent ~ 23 ps. Crucially, the data do not exhibit a measured plateau: the resolution is still falling, if gently, at the brightest events, so the floor is an extrapolation and its true uncertainty exceeds the fit error. Three estimates nonetheless converge: the amplitude fit (24.7 ps), the energy fit (~ 23 ps), and the physical expectation from shower-depth fluctuation discussed in Section 6 (~ 28 ps, i.e. $\sim 1 X_0$). The convergence on ~ 23 –28 ps makes a substantially lower floor unlikely.

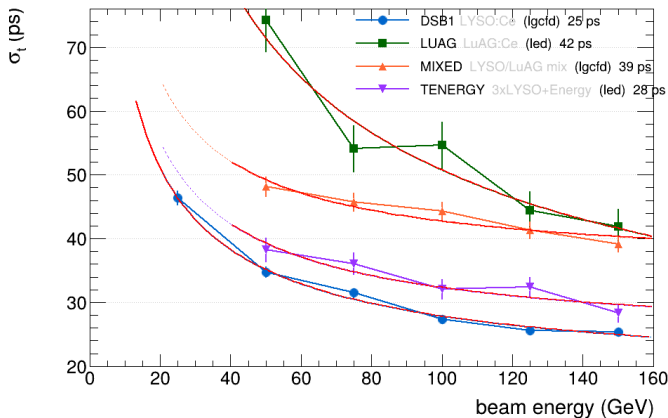


Figure 7: Best-achievable $(\text{DW} - \text{UP})/2$ resolution versus energy for the four builds. The ordering follows the scintillation light yield: high-light LYSO (DSB1) lowest, low-light LuAG highest.

Table 1: Best-achievable resolution (brightest-1000 slice) and extrapolated slew floor at 150 GeV for the four builds.

Build	σ_t (brightest)	Slew floor b
DSB1 (LYSO, high light)	25.4 ps	24.7 ± 1.1 ps
TENERGY (3 $\times$ LYSO+E)	28.7 ps	28.1 ± 1.6 ps
MIXED (LYSO+LuAG)	40.0 ps	40.2 ± 0.9 ps
LUAG (LuAG, low light)	41.2 ps	36.0 ± 3.0 ps

The earlier $\sigma_t = a/\sqrt{E} \oplus b$ parametrisation of Ref. [2] returns a lower floor, $b = 17.5$ ps. That form assumes a photostatistics $(1/\sqrt{E})$ stochastic term; the data of Section 5.2, however, fall as $1/\Sigma\text{LG}$ (slew-rate limited), more steeply than $1/\sqrt{E}$. A $1/\sqrt{E}$ fit forced through such data compensates with an artificially low floor. The slew interpretation, which matches the observed scaling, assigns more of the measured resolution to the irreducible term and raises the floor to ~ 25 ps, identifying it with shower-depth fluctuation (Section 6).

5.5. Cross-build comparison: light yield sets the clock

Applying the same analysis to all four builds (Fig. 7) orders them cleanly by scintillation light yield. The best-achievable (brightest-slice) resolution at 150 GeV is 25.4 ps for high-light LYSO (DSB1), 28.7 ps for the LYSO+energy build (TENERGY), 40.0 ps for the mixed build (MIXED), and 41.2 ps for low-light LuAG (LUAG); the corresponding slew floors are 24.7, 28.1, 40.2 and 36.0 ps. More light gives a faster edge and better timing, exactly as the slew argument predicts. (For the low-light builds a fixed-threshold leading-edge discriminator is the most robust timing source, the LG-ratio CFD being susceptible to the small pulses; for the LYSO-rich builds hg_lgcfid is used.)

6. Discussion

The measurements show that, with the rising edge recovered, RADICAL timing is slew-rate limited over the entire accessible range: $\sigma_t \propto 1/\Sigma\text{LG}$, and the resolution is a function of collected light alone. At the brightest events the slew term has fallen to only ~ 5 ps and the floor dominates, so additional light yields rapidly diminishing returns — a doubling of light would improve σ_t by only a few tenths of a picosecond.

What, then, is the floor? It cannot be electronics jitter alone, which the four-channel-per-end average already suppresses. The differential estimator $(\text{DW} - \text{UP})/2$ is, by construction, sensitive to the longitudinal position of the shower: light from a deeper shower reaches the downstream end earlier and the upstream end later, shifting the two in opposite senses. Shower-to-shower fluctuation in the depth of energy deposition therefore produces an irreducible spread in $t_{\text{DW-UP}}$ that does not decrease with light, because it is common to all channels at an end and geometric rather than statistical in origin. Its scale, $\sim 1 X_0$ of path-length difference, is ~ 25 – 28 ps, matching the fitted floor. The very feature that makes the estimator MCP-free — its differential construction — is also what couples it to shower depth.

This identification points to the route below the floor. The same channel-to-channel light-sharing that carries the depth information could be used to estimate each shower's depth and subtract its timing bias event by event. Such a depth-corrected estimator is the natural next step and is the subject of ongoing work.

7. Conclusions

We have demonstrated that saturation of the timing front end in a W/LYSO sampling calorimeter need not limit its time resolution. By predicting the unclipped pulse peak from a linear low-gain copy and timing a constant fraction of that predicted peak, the steep rising edge lost to clipping is recovered, improving on the clipped-peak result. Using an MCP-free differential estimator we measure a best-achievable resolution of 25.4 ps for 150 GeV electrons in the high-light LYSO build, and we show that the resolution is slew-rate limited, scaling as the inverse of the collected light and depending on shower brightness rather than beam energy. The resolution extrapolates to a floor of ≈ 25 ps, consistent across amplitude- and energy-based extrapolations and with the expected shower-depth fluctuation of $\sim 1 X_0$. Across four builds the resolution tracks the scintillation light yield. The dominant remaining limitation is the shower itself; a depth-corrected estimator offers a path beyond it.

Acknowledgements

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References

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